

SUMMARY

- Electrical Engineer with proven expertise in power electronics validation, data analytics, and embedded systems, with a focus on DC fast chargers, bi-directional inverters, and DC-DC converters.
- Proficient in Python, MATLAB, C/C++, and simulation tools like PLECS, LTspice, and Simulink for modeling control algorithms and analyzing system performance.
- Applied statistical analysis and signal processing to large datasets for predictive diagnostics and reliability improvements in high-voltage systems.
- Experienced in developing and executing V&V test plans, conducting EMI/EMC testing, and performing DFMEA and fault tree analysis to ensure compliance with UL, ISO, and IEC standards.
- Skilled in PLC programming (Siemens, Allen-Bradley) and hardware-in-the-loop (HIL) setups for automated testing and control system validation.
- Strong foundation in embedded systems, firmware integration, and cross-functional collaboration across mechanical, PCB, and controls teams.
- Demonstrated ability to manage projects independently within Agile teams, ensuring timely delivery of robust, scalable engineering solutions.

SKILLS

Power Electronics & Charging Systems	DC-DC Converters, AC-DC Inverters, DC fast chargers, Bi-directional Inverters, Magnetics, EMI/EMC, PMICs, Battery management systems
Testing & Validation	V&V Test Plans, DFMEA, Fault Tree Analysis, Hazard Analysis, High-Voltage Testing, Functional safety testing
Lab Equipment	Oscilloscopes, Power Analyzers, Thermal Cameras, DAQ Systems
Simulation & Modeling	MATLAB/Simulink, PLECS, LTspice, Control Algorithm Modeling
Data Analytics & Processing	Power BI, Streamlit, Pandas, NumPy, PIL, Matplotlib, Statistical Analysis, Signal Processing
Automation & PLC Systems	Ladder Logic, Structured Text, Siemens, Allen-Bradley, HIL Systems
Embedded Systems	Microcontroller Programming, Firmware Integration, Real-Time Control Systems
Product Development	Prototyping, Debugging, DFM/DFT, Compliance (UL, ISO, IEC)
Project Management and Collaboration	Agile (Scrum, Kanban), Slack, Microsoft Teams, Zoom, Google Workspace, Microsoft Office
Programming Language	Python, SQL, MATLAB, C/C++
Software & Tools	Git, GitHub, Version Control, PCB Design Collaboration, CAD Tools

PROFESSIONAL EXPERIENCE

- Data Analytics & Validation Engineer | CAS Dataloggers | Chesterland, OH

Feb 2025 – Present
- Lead design validation and performance testing of advanced power electronics systems, including Active Front-End (AFE) converters, bi-directional inverters, and DC-DC converters, ensuring compliance with UL, ISO, and IEC standards.
 - Perform simulation and analysis of inverter and switching circuits using MATLAB/Simulink, PLECS, and LTspice, evaluating efficiency, control stability, and thermal performance under varying load and temperature conditions.
 - Develop and optimize motor drive systems by analyzing IPM motor characteristics, implementing FOC and PWM control methods, and verifying performance through HIL and bench-level testing.
 - Conduct thermal modeling and characterization of power modules, MOSFETs, and IGBTs, applying accurate junction temperature calculations and loss analysis to improve reliability and derating strategies.
 - Drive hardware design, prototyping, and validation of inverter stages and gate driver circuits, collaborating with PCB layout, firmware, and controls teams for full system integration.
 - Utilize measurement and diagnostic instrumentation—including oscilloscopes, power analyzers, and thermal cameras—to execute fault analysis, switching characterization, and system efficiency mapping.

- Develop automated data acquisition and analytics frameworks using Python and MATLAB, enabling real-time test monitoring, performance visualization, and predictive diagnostics.
- Research and implement Hardware-in-the-Loop (HIL) simulations for converter control validation, integrating PLC-based automation (Siemens, Allen-Bradley) for test sequencing and closed-loop verification.
- Perform root cause and reliability analysis on inverter subsystems to guide design improvements in thermal management, switching performance, and EMI/EMC compliance.
- Collaborate with cross-functional engineering teams and suppliers to ensure design for manufacturability (DFM) and design for testability (DFT), supporting robust production deployment.

Telecommunication R&D – Intern | Bharti Airtel | India **2022**

Jan 2022 – May

- Conducted data-driven performance analysis on large-scale network traffic and base station logs using Python and MATLAB, identifying system bottlenecks, and proposing optimization strategies applicable to real-time power and control systems.
- Developed and tested control system algorithms to enhance network uptime and operational efficiency, contributing to real-time decision-making frameworks like those used in industrial power conversion systems.
- Built simulation models for communication protocols and adaptive control mechanisms, supporting predictive analytics, and improving system reliability—skills transferable to inverter/converter modeling and validation.
- Gained hands-on experience with embedded systems and hardware testing, integrating control logic and validating system behavior under dynamic conditions, including fault scenarios and recovery strategies.
- Collaborated with cross-functional engineering teams to troubleshoot system failures, applying structured root cause analysis and implementing corrective actions aligned with safety and reliability standards.
- Supported automation and connectivity initiatives, gaining foundational experience in machine-to-machine (M2M) communication, relevant to industrial automation, PLC integration, and smart charging systems.
- Contributed to version-controlled development using Git, maintaining clean, traceable code, and collaborating on shared repositories for algorithm development and system simulation.

ACADEMIC PROJECT

Design and Simulation of a Bi-Directional DC-DC Converter for EV Applications

- Modeled and simulated a bi-directional converter using MATLAB/Simulink and PLECS for electric vehicle battery charging and regenerative braking.
- Performed efficiency analysis, thermal modeling, and control loop tuning.
- Validated performance under varying load and temperature conditions.

Predictive Maintenance Using Data Analytics for Electrical Systems

- Analyzed historical sensor data from simulated electrical systems using Python, Pandas, and Scikit-learn.
- Built a machine learning model to predict component failures based on temperature, current, and voltage trends.
- Achieved 85% accuracy in early fault detection, improving system reliability.

Control Algorithm Development for Grid-Tied Inverter

- Developed and simulated MPPT and voltage regulation algorithms for a grid-tied solar inverter.
- Used MATLAB/Simulink to model inverter dynamics and validate control strategies.
- Evaluated system performance under varying solar irradiance and load conditions.

EDUCATION

Master of Science in Electrical and Electronics Engineering, Stevens Institute of Technology

Dec 2024

Bachelor of Technology in Electronics and Communication Engineering, Nirma University

Jun 2022